



MIST- MEDICAL INSTITUTE FOR SCREENING TEST

PHYSIOLOGY SUBJECT WISE TEST DATED ON 29th MAY 2026

QUESTIONS

- Q1. As part of a teaching demonstration, body-fluid compartments in a 50-year-old male are estimated. After equilibration, a total body water marker gives a volume of 42 L. A labelled inulin marker gives an extracellular fluid volume of 14 L. Evans blue dye estimates plasma volume as 3.5 L. No tracer loss occurs during the equilibration period. Which calculated pair is most appropriate?
- a) ICF = 28 L; interstitial fluid = 10.5 L
 - b) ICF = 14 L; interstitial fluid = 28 L
 - c) ICF = 10.5 L; interstitial fluid = 3.5 L
 - d) ICF = 42 L; interstitial fluid = 14 L
- Q2. A 26-year-old woman with psychogenic polydipsia is brought to the emergency department. She drank nearly 8 liters of water over several hours. She develops headache, vomiting, confusion, and a generalized seizure. Serum sodium is 116 mEq/L, glucose is normal, and BUN is normal. Measured plasma osmolality is low. CT brain shows early cerebral edema. There is no history of diarrhea, diuretic use, or blood loss. Which fluid-compartment change best explains her neurologic symptoms?
- a) ECF osmolarity rises, causing water to leave neurons
 - b) ECF osmolarity falls, causing water to enter neurons
 - c) ECF volume contracts without change in ICF volume
 - d) ICF volume contracts due to increased extracellular sodium
- Q3. A 49-year-old man with long-standing alcoholic cirrhosis presents with ankle edema and ascites. His blood pressure is normal. Serum albumin is 1.8 g/dL. Urinalysis shows no proteinuria. Diagnostic paracentesis shows a low-protein transudative fluid. There is no evidence of cellulitis or increased capillary permeability. A student asks which Starling force has changed most directly. Which variable is primarily responsible for the edema?
- a) Increased plasma oncotic pressure
 - b) Decreased plasma oncotic pressure
 - c) Decreased capillary filtration coefficient
 - d) Increased osmotic reflection coefficient
- Q4. A 68-year-old man with systolic heart failure is started on digoxin. During rounds, a resident explains its positive inotropic effect. Which membrane transport process is directly inhibited by digoxin?



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- a) Secondary active Na^+ /glucose cotransport
 - b) Facilitated diffusion through GLUT transporters
 - c) Primary active transport by Na^+/K^+ -ATPase
 - d) Simple diffusion of potassium through leak channels
- Q5. A 61-year-old man with missed dialysis presents with muscle weakness and palpitations. ECG shows tall peaked T waves. Serum potassium is 7.2 mEq/L. The professor asks why severe hyperkalemia can paradoxically reduce excitability. Which explanation is most accurate?
- a) Increased extracellular K^+ makes the K^+ equilibrium potential more negative
 - b) Increased extracellular K^+ makes the resting membrane potential less negative and can inactivate Na^+ channels
 - c) Increased extracellular K^+ hyperpolarizes skeletal muscle fibers
 - d) Increased extracellular K^+ abolishes potassium permeability at rest
- Q6. An anterolateral cordotomy is performed that produces relief of pain in the right leg. It is effective because it interrupts which of the following?
- a) Left dorsal column
 - b) Left anterolateral spinothalamic tract
 - c) Right dorsal column
 - d) Right anterolateral spinothalamic tract
- Q7. The representational (or non-dominant) hemisphere is better than the categorical (or dominant) hemisphere in which of the following?
- a) Speech
 - b) Recognition of objects by their form
 - c) Mathematical calculations
 - d) Understanding printed and spoken words
- Q8. A 75-year-old lady presents with vascular injury to the inferior frontal gyrus. Which functional area is most likely to be affected?
- a) Visual
 - b) Auditory
 - c) Wernicke's



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- d) Broca's
- Q9. A newborn develops severe watery diarrhea after feeds containing glucose and galactose. The consultant asks how apical glucose uptake is transported into the cell. Which is the best answer?
- a) Simple diffusion
 - b) Primary active transport
 - c) Secondary active cotransport
 - d) Facilitated diffusion
- Q10. A 24-year-old man presents with ascending weakness 2 weeks after an episode of gastroenteritis. Examination shows areflexia, impaired vibration sense, and symmetric limb weakness. Nerve conduction study shows reduced conduction velocity. Sensory nerve action potentials are also slowed. The neurologist explains that the primary abnormality is not loss of axons but loss of insulation. A physiology resident is asked which property of the nerve fiber is most affected. Which mechanism best explains the slowed conduction?
- a) Reduced axonal diameter causing reduced axial resistance
 - b) Loss of myelin causing failure of rapid saltatory conduction
 - c) Increased Na^+/K^+ -ATPase activity at internodes
 - d) Increased number of functioning axons in the nerve trunk
- Q11. A patient with history of head injury following road traffic accident is on treatment for his inability to sleep. The patient asks the doctor the reason behind his inability to sleep. The likely cause is which of the following?
- a) Overactive amygdala
 - b) Inactive amygdala
 - c) Overactive reticular system
 - d) Inactive reticular system
- Q12. Before injection, stimulation of the mandibular nerve produces sharp pain. After lignocaine infiltration, the same stimulus no longer produces pain sensation. The nerve membrane remains structurally intact. The anesthetic is most effective when the nerve is repeatedly active. Electrophysiology shows failure of the rapid upstroke of the action potential. Resting membrane potential is nearly unchanged. Which ion-channel event is primarily blocked?
- a) Opening of voltage-gated Na^+ channels during depolarization



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- b) Opening of voltage-gated K^+ channels during repolarization
- c) Passive K^+ leak through resting channels
- d) Acetylcholine binding to nicotinic receptors at the motor end plate

Q13. A 36-year-old woman complains of diplopia and difficulty keeping her eyes open by evening. She also reports nasal speech after prolonged conversation. Symptoms improve after rest. Examination shows fatigable ptosis and mild proximal limb weakness. Repetitive nerve stimulation shows a decremental muscle response. Sensory examination and deep tendon reflexes are normal. Serum testing reveals antibodies against a postsynaptic motor end-plate protein. Which physiological abnormality best explains her weakness?

- a) Reduced presynaptic Ca^{2+} influx through P/Q-type channels
- b) Reduced number of functional nicotinic acetylcholine receptors
- c) Excess acetylcholine release from motor nerve terminals
- d) Defective ryanodine receptor opening in sarcoplasmic reticulum

Q14. A 41-year-old man develops blurred vision, diplopia, dysphagia, and descending weakness. He had eaten improperly preserved home-canned food the previous day. Sensory examination is normal. Deep tendon reflexes are reduced. The ICU team suspects a toxin-mediated neuromuscular junction disorder. Which molecular event best explains the paralysis?

- a) Competitive blockade of postsynaptic nicotinic receptors
- b) Inhibition of acetylcholinesterase at the synaptic cleft
- c) Cleavage of SNARE proteins preventing acetylcholine vesicle fusion
- d) Increased acetylcholine release causing sustained tetanic contraction

Q15. A 25-year-old patient after head trauma presented to the Emergency room. On neurological examination, he was found to have normal handwriting and could copy shapes of objects but could not speak. Localize the CNS lesion of this patient:

- a) Wernicke's area
- b) Broca's area
- c) Arcus fasciculus
- d) Internal capsule

Q16. A 59-year-old chronic smoker presents with progressive proximal leg weakness. He says he feels slightly stronger after walking a few steps. Deep tendon reflexes are reduced at rest. Repeated



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brief exercise transiently improves muscle strength. Repetitive nerve stimulation at high frequency shows an incremental response. Chest imaging later reveals a small-cell lung carcinoma. Which abnormality is most likely responsible?

- a) Autoantibodies against postsynaptic nicotinic acetylcholine receptors
- b) Autoantibodies against presynaptic voltage-gated Ca^{2+} channels
- c) Cleavage of SNARE proteins by botulinum toxin
- d) Defective dystrophin anchoring of sarcolemma

Q17. A cross-sectional view of a skeletal muscle fiber through the H-zone would reveal the presence of which of the following?

- a) Actin and titin
- b) Actin and myosin
- c) Myosin only
- d) Actin only

Q18. A person exposed to high temperatures while exercising vigorously. To maintain internal body temperature within a narrow range, which physiological response most likely to occur?

- a) Sweating and vasodilation in skin
- b) Shivering and vasoconstriction in skin
- c) Increased production of heat by liver
- d) Decreased heart rate and respiratory rate

Q19. A 45-year-old female patient presents with intermittent, sudden, violent, involuntary movement involving the right arm is due to a lesion in which of the following?

- a) Substantia nigra
- b) Subthalamic nucleus
- c) Globus Pallidus
- d) Caudate nucleus

Q20. A patient having unilateral drooping of eyelid, muscle weakness, headache. Symptoms worsen during evening. Patient feels better while taking neostigmine. What is the most probable cause of the patient's condition?



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- a) Lambert Eaton syndrome
- b) Ophthalmoplegia
- c) Myasthenia gravis
- d) Becker's muscular dystrophy

Q21. The lung volume at the end of a normal, quiet expiration is which of the following?

- a) Inspiratory capacity
- b) Forced vital capacity
- c) Functional residual capacity
- d) Total lung capacity

Q22. An experimental animal (monkey) develops increased sexual activity, increased appetite, anger and no fear following bilateral removal of a certain area of brain. Removal of which of the following structures is responsible for the mentioned changes in behavior?

- a) Amygdala
- b) Thalamus
- c) Cingulate gyrus
- d) Hypothalamus

Q23. Administration of i/v norepinephrine causes a reflex decrease in heart rate. What is the most likely physiological mechanism that is responsible for this response?

- a) Stimulation of the baroreceptors
- b) Stimulation of joint proprioceptors
- c) Bezold- Jarisch reflex
- d) Stimulation of chemoreceptors

Q24. On i/v injection of an irritant substance, bradycardia, hypotension and apnea developed in the subject. This represents which of the following reflex mechanism?

- a) Bezold – Jarisch Reflex
- b) Brainbridge Reflex



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- c) Cushing's Reflex
- d) J-receptor reflex

Q25. In the JVP, the "a" wave is due to which of the following?

- a) Tricuspid bulging into the right atrium
- b) Right atrial relaxation
- c) Right atrial filling
- d) Right atrial contraction

Q26. A 59-year-old man with advanced chronic kidney disease has fatigue and normocytic anemia. Iron studies are adequate. There is no active bleeding. The nephrologist explains that diseased renal tissue produces insufficient erythropoietin. Which ONE statement regarding erythropoietin is false?

- a) In adults, most erythropoietin is produced by cells in the renal cortex.
- b) Erythropoietin stimulates red-cell production in bone marrow.
- c) Reduced oxygen availability can stimulate erythropoietin production.
- d) Chronic kidney disease anemia is usually caused by excess erythropoietin production suppressing erythropoiesis.

Q27. Which of the following causes a rightward shift in the oxygen-hemoglobin dissociation curve?

- a) Increase in PCO₂
- b) Decrease in 2,3-BPG
- c) Decrease in PCO₂ (hypocapnia)
- d) Decrease in temperature

Q28. A 42-year-old woman undergoes surgery for a suprasellar mass. The pituitary stalk is damaged. Postoperatively, TSH, FSH/LH, and GH are reduced. However, prolactin rises. The endocrinologist explains that most anterior pituitary hormones require hypothalamic releasing hormones. Prolactin is different because it is under tonic hypothalamic inhibition. Which ONE statement is false?

- a) Stalk interruption reduces delivery of releasing hormones to the anterior pituitary.
- b) TSH, gonadotropins, and GH may fall after stalk section.
- c) Dopamine normally inhibits prolactin secretion.



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d) Prolactin falls after stalk section because dopamine delivery to lactotrophs increases.

Q29. A 19-year-old man undergoes an oral glucose tolerance test. Glucose enters pancreatic β -cells and is metabolized. The ATP/ADP ratio rises. An ion channel closes, reducing potassium efflux. The β -cell depolarizes, voltage-gated calcium channels open, and insulin granules are exocytosed. A pharmacology resident notes that sulfonylureas exploit the same channel pathway. Which ONE statement is false?

- a) Glucose metabolism raises β -cell ATP/ADP ratio.
- b) ATP-sensitive potassium-channel closure depolarizes β -cells.
- c) Calcium influx triggers insulin granule exocytosis.
- d) Glucose-mediated insulin release is primarily initiated by insulin-receptor phosphorylation on β -cells.

Q30. A 35-year-old woman presents with vomiting, abdominal pain, fever, and confusion. BP is 78/44 mmHg, glucose 52 mg/dL, sodium 121 mEq/L, and potassium 6.0 mEq/L. She has hyperpigmentation of palmar creases and buccal mucosa. Morning cortisol and aldosterone are low. ACTH and renin are high. She improves with IV hydrocortisone, saline, and dextrose. Which ONE statement is false?

- a) Primary adrenal insufficiency can cause low cortisol, low aldosterone, high ACTH, and high renin.
- b) Cortisol deficiency can impair vascular responsiveness to catecholamines.
- c) Hyperpigmentation occurs because ACTH is suppressed in primary adrenal failure.
- d) Hyponatremia may reflect mineralocorticoid deficiency and ADH-mediated water retention.

Q31. A child with cystic fibrosis has bulky foul-smelling stools and poor weight gain. Stool elastase is low. In a teaching session, duodenal acid infusion increases bicarbonate-rich pancreatic secretion. A fatty protein meal increases enzyme-rich secretion. Which ONE statement is false?

- a) Secretin is released by duodenal acid and stimulates pancreatic ductal bicarbonate secretion.
- b) CCK is released mainly by fat and amino acids and stimulates pancreatic enzyme secretion.
- c) Pancreatic proteases are secreted as inactive zymogens.
- d) CCK is released primarily by acid and is the dominant hormone for ductal bicarbonate secretion.

Q32. A 50-year-old patient presents with respiratory difficulty and chronic smoking for the last 30 years. Pulmonary function testing shows a decrease in FEV1%. What is the most probable diagnosis?



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- a) Chronic Obstructive Pulmonary Disease (COPD)
- b) Pulmonary fibrosis
- c) Interstitial lung disease
- d) Ca lung

Q33. A volunteer has urine flow 2 mL/min. Plasma inulin is 0.5 mg/dL, urine inulin is 30 mg/dL. Plasma PAH is 0.5 mg/dL, urine PAH is 150 mg/dL. Hematocrit is 40%. PAH dose is low enough that secretion is unsaturated. The examiner later asks what happens when plasma PAH is raised very high. Which ONE statement is false?

- a) Inulin clearance is approximately 120 mL/min.
- b) PAH clearance is approximately 600 mL/min, estimating effective renal plasma flow.
- c) Renal blood flow is approximately 1000 mL/min.
- d) At very high plasma PAH, clearance exceeds true renal plasma flow because extraction becomes greater than 100%.

Q34. Two medical students are planning a trip to Leh (high altitude). They want to study the effects of acclimatization in high altitude. Which one of the following changes will not be seen in high altitude during the process of acclimatization?

- a) Increase in RBC count
- b) Increase in capillary density
- c) Increase in renal excretion of bicarbonates
- d) Hypoventilation

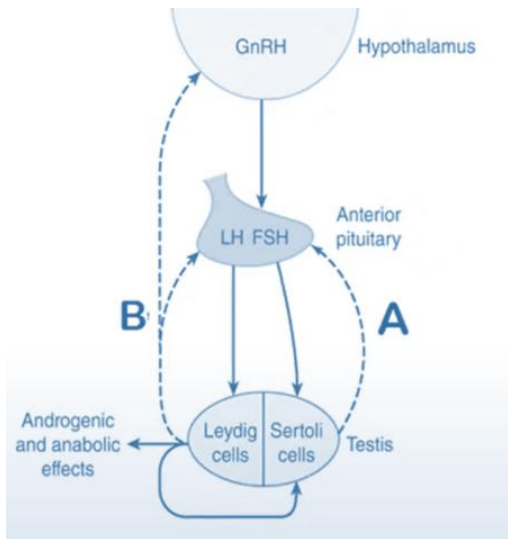
Q35. Action of ANP (atrial natriuretic peptide) is mediated by which of the following?

- a) cAMP
- b) cGMP
- c) IP3-DAG
- d) JAK/STAT

Q36. Identify the hormones A & B responsible for negative feedback on Hypothalamus – Pituitary axis?



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- a) Inhibin & Testosterone
- b) Testosterone & Inhibin
- c) DHA & Testosterone
- d) Testosterone and DHA

Q37. A 58-year-old man with pulmonary edema receives IV furosemide. Urine output rises markedly. Despite high endogenous ADH, his ability to concentrate urine is reduced. Serum potassium falls and bicarbonate rises. Urinary calcium and magnesium excretion increase. Which ONE statement is false?

- a) Loop diuretics inhibit NKCC2 in the thick ascending limb.
- b) Loop diuretics impair the medullary gradient and reduce concentrating ability.
- c) Increased distal sodium delivery can promote potassium and hydrogen secretion.
- d) Loop diuretics strengthen the lumen-positive voltage and decrease urinary calcium loss.

Q38. A 41-year-old man is found confused in a closed garage. His wife also has headache and nausea. ABG shows PaO₂ 96 mmHg, but co-oximetry shows carboxyhemoglobin 32%. Lactate is elevated. Conventional pulse oximetry reads 98%. He is treated with 100% oxygen, and hyperbaric oxygen. Which ONE statement is false?

- a) PaO₂ may be normal because it reflects dissolved oxygen in plasma.
- b) CO reduces arterial oxygen content by occupying hemoglobin binding sites.
- c) CO shifts the oxyhemoglobin dissociation curve leftward, impairing tissue unloading.
- d) Standard two-wavelength pulse oximetry reliably distinguishes oxyhemoglobin from carboxyhemoglobin.



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- Q39. Which of the following causes the largest increase in osmolarity of plasma?
- a) Glucose
 - b) Plasma protein
 - c) Sodium
 - d) Bicarbonate
- Q40. A 40-year-old patient has unilateral drooping of eyelid, muscle weakness, headache. Symptoms worsen during evening. Patient feels better while taking neostigmine. What is the most probable cause?
- a) Lambert- Eaton syndrome
 - b) Ophthalmoplegia
 - c) Myasthenia gravis
 - d) Becker's muscular dystrophy
- Q41. Large 'v' wave on JVP seen in which of the following?
- a) Tricuspid regurgitation
 - b) Aortic regurgitation
 - c) Atrial fibrillation
 - d) Tricuspid stenosis
- Q42. Which of the following statements is true about GFR?
- a) Increase in renal plasma flow (RPF) increases GFR
 - b) Increased sympathetic function leads to increased GFR
 - c) Afferent arterioles constriction results in increased GFR
 - d) Decrease in RPF increases GFR
- Q43. A 27-year-old climber rapidly ascends to 4,500 meters. He develops headache, insomnia, tachycardia, and hyperventilation. ABG shows low PaO_2 , low PaCO_2 , and alkalemia. After several days, serum bicarbonate falls and ventilation becomes more sustained. Hematocrit begins to rise over days to weeks. Another climber develops high-altitude pulmonary edema. Which ONE statement is false?
- a) Low barometric pressure lowers inspired and alveolar PO_2 .



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- b) Carotid-body stimulation increases ventilation and lowers PaCO_2 .
- c) Renal compensation consists mainly of bicarbonate retention to restore PaCO_2 to sea-level values.
- d) Hypoxic pulmonary vasoconstriction can increase pulmonary arterial pressure.
- Q44. If a person starts consuming more sodium suddenly, which of the following statements is correct?
- a) More intake of Na^+ causes more excretion of Na^+
- b) More intake of Na^+ causes less excretion of Na^+
- c) More intake of Na^+ decreases thirst
- d) More intake of Na^+ decreases ECF
- Q45. A premature infant born at 28 weeks develops tachypnea, retractions, grunting, and hypoxemia. Chest radiograph shows low lung volumes and diffuse reticulogranular opacities. CPAP improves oxygenation. Exogenous surfactant is administered. The neonatologist explains that small alveoli collapse when surface tension is high. Which of the following statements regarding surfactant is false?
- a) Surfactant lowers alveolar surface tension.
- b) Surfactant increases lung compliance and reduces work of breathing.
- c) Without surfactant, smaller alveoli require higher pressure to remain open.
- d) Surfactant deficiency decreases surface tension and prevents atelectasis.
- Q46. Onset of which of the following phases of cardiac cycle corresponds to the first heart sound?
- a) Aortic ejection
- b) Isovolumetric contraction
- c) Diastasis
- d) Rapid atrial filling
- Q47. A 69-year-old man with severe COPD presents with fever, wheeze, and somnolence. Room-air ABG shows pH 7.34, PaCO_2 58 mmHg, HCO_3^- 31 mEq/L, and PaO_2 47 mmHg. He is placed on uncontrolled high-flow oxygen. Two hours later, PaO_2 is very high but PaCO_2 rises to 78 mmHg. The pulmonologist explains that oxygen was necessary but excessive oxygen can worsen CO_2 retention. Which of the following statements is false?
- a) Elevated bicarbonate suggests chronic renal compensation for respiratory acidosis.



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- b) High inspired oxygen can worsen V/Q mismatch by reversing hypoxic pulmonary vasoconstriction.
- c) Oxygenation of hemoglobin reduces its CO₂-carrying capacity through the Haldane effect.
- d) Central chemoreceptors directly sense arterial oxygen tension and stop firing when PaO₂ normalizes.

Q48. Calmodulin is most closely related, both structurally and functionally, to which of the following proteins?

- a) G-actin
- b) Myosin light chain
- c) Tropomyosin
- d) Troponin-C

Q49. A child was defecating in the open in a village when he was attacked by pack of dogs. One of the attacking dogs was found dead the next day. Considering the high risk of rabies in the child he was admitted to District hospital and was given antirabies serum (ARS) and Rabipur vaccine. Which of the following is responsible for transport of rabies virus to brain?

- a) Dynein
- b) Kinesin
- c) Actin
- d) Vimentin

Q50. The reward centre of the hypothalamus is which of the following?

- a) Lateral hypothalamus
- b) Ventromedial hypothalamus
- c) Preoptic nucleus
- d) Posterior hypothalamus

Q51. A neuroscientist is studying communication between the hypothalamus and pituitary in a rat model. She interrupts blood flow through the median eminence and then measures circulating levels of pituitary hormones following appropriate physiologic stimulation. Secretion of which of the following hormones will be unaffected by the experimental manipulation?

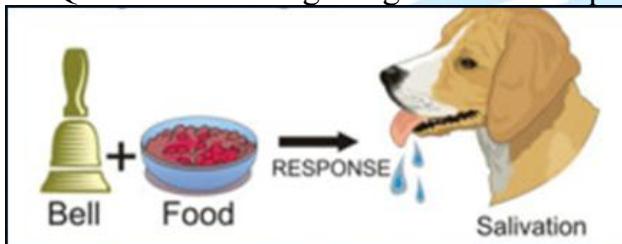
- a) Growth hormone



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- b) Prolactin
- c) Thyroid-stimulating hormone
- d) Vasopressin

Q52. The diagram given below represents which of the following?



- a) Conditioned response
 - b) Unconditioned response
 - c) Classical conditioning
 - d) Operant conditioning
- Q53. All of the following are types of implicit memory except?
- a) Semantic memory
 - b) Procedural memory
 - c) Associative memory
 - d) Non associative memory
- Q54. Which extrapyramidal tract functions like the pyramidal tract?
- a) Anterior corticospinal
 - b) Tectospinal
 - c) Rubrospinal
 - d) Vestibulospinal
- Q55. The thalamic relay nucleus for taste is which of the following?
- a) Ventroposteromedial
 - b) Ventroposterolateral
 - c) Mediodorsothalamic



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d) Lateral geniculate body

Q56. The major secretion of the adrenal medulla is which of the following?

- a) Nor epinephrine
- b) Epinephrine
- c) Dopamine
- d) Adrenomedulin

Q57. A 25-year-old man meets with a road traffic accident that completely eliminates all nerve traffic in the right half of the spinal cord at C2. Loss of which of the following is likely in the right hand of this man?

- a) Crude touch and pain sensation
- b) Crude touch and temperature sensation
- c) Motor function and temperature sensation
- d) Motor function and vibration sense

Q58. Which of the following aquaporins is involved in water reabsorption from the PCT?

- a) Aquaporin 1
- b) Aquaporin 2
- c) Aquaporin 3
- d) Aquaporin 4

Q59. A 24-year-old woman presents with sudden palpitations at 190/min. ECG shows a regular narrow-complex tachycardia consistent with AV nodal reentry. Carotid massage transiently slows the rhythm. Adenosine terminates the tachycardia. Which of the following statements regarding pacemaker potentials is incorrect?

- a) SA and AV nodal phase 0 depends mainly on L-type calcium current.
- b) Vagal stimulation increases potassium conductance and slows nodal firing.
- c) Sympathetic β_1 stimulation steepens phase 4 depolarization.
- d) SA and AV nodal phase 0 depends mainly on fast voltage-gated sodium channels.

Q60. Under normal conditions, maximum calcium reabsorption occurs in which of the following segments of the nephron?



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- a) PCT
- b) TAL
- c) DCT
- d) CD

Q61. Which of the following GI secretions has the highest pH?

- a) Saliva
- b) Succus entericus
- c) Pancreatic juice
- d) Brunners' gland secretion

Q62. A 39-year-old woman develops hypotension after a central venous catheter attempt. She has tachycardia, muffled heart sounds, elevated JVP, and pulsus paradoxus. Echo shows a large pericardial effusion with right atrial collapse. The JVP tracing has prominent x descent and blunted y descent. Urgent pericardiocentesis improves blood pressure. Which of the following statements is false?

- a) Tamponade limits diastolic filling by increasing pericardial pressure.
- b) Pulsus paradoxus reflects exaggerated inspiratory reduction in LV filling and stroke volume.
- c) Blunted y descent reflects impaired rapid early diastolic filling.
- d) Pericardiocentesis improves cardiac output primarily by reducing systemic vascular resistance.

Q63. Which of the following tracts is mainly responsible for fine, skilled movements?

- a) Lateral corticospinal tract
- b) Vestibulospinal tract
- c) Reticulospinal tract
- d) Rubrospinal tract

Q64. Golgi tendon organ senses which of the following?

- a) Muscle tension
- b) Static muscle length
- c) Velocity of change of muscle length
- d) Pain during stretch



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- Q65. Which hormone is required to cross the plasma membrane for its actions?
- a) Epinephrine
 - b) Thyroxine
 - c) Glucagon
 - d) Insulin
- Q66. The lung volume at the end of a maximum inspiration is :
- a) Inspiratory capacity
 - b) Forced vital capacity
 - c) Functional residual capacity
 - d) Total lung capacity
- Q67. Cardiac index is which of the following?
- a) Cardiac output per kg body weight
 - b) Cardiac output per meter height
 - c) Cardiac output per sq m of body surface area
 - d) Cardiac output per unit mass
- Q68. Increase in sympathetic stimulation causes all of the following except?
- a) Increase in heart rate
 - b) Increase in force of contraction of the heart
 - c) Increase in peripheral vascular resistance
 - d) Increase in venous compliance
- Q69. A 76-year-old man has exertional syncope and angina. Carotid upstroke is delayed and low amplitude. There is a harsh systolic murmur at the right upper sternal border radiating to the carotids. Echo shows severe calcific aortic stenosis and concentric LV hypertrophy. LV systolic pressure exceeds aortic systolic pressure during ejection. The examiner compares this with chronic aortic regurgitation. Which ONE statement is false?
- a) Aortic stenosis causes pressure overload and concentric LV hypertrophy.
 - b) Aortic stenosis can cause exertional syncope because cardiac output cannot rise adequately during exercise
 - c) Aortic stenosis classically produces bounding pulses and wide pulse pressure due to diastolic



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runoff.

- d) Reduced preload usually softens the murmur of fixed valvular aortic stenosis.

Q70. Which of the following is true regarding surfactant lining the alveoli?

- a) Surfactant helps prevent alveolar collapse.
- b) Surfactant is produced in alveolar type I cells and secreted into the alveolus.
- c) Surfactant is increased in the lungs of heavy smokers.
- d) Surfactant is a glycolipid complex.

Q71. Which of the following occurs in high altitude?

- a) Hypoxic hypoxia
- b) Anemic hypoxia
- c) Stagnant hypoxia
- d) Histotoxic hypoxia

Q72. A 72-year-old man with long-standing diabetes feels dizzy when standing. Supine BP is 144/78 mmHg, HR 74/min. After standing, BP falls to 88/54 mmHg, but HR rises only to 78/min. He has erectile dysfunction, reduced sweating, and distal neuropathy. There is no bleeding or fever. Which of the following statements is incorrect?

- a) Standing causes venous pooling and reduces venous return.
- b) Reduced stroke volume lowers pulse pressure before reflex compensation.
- c) Normal compensation includes sympathetic vasoconstriction and reduced vagal tone.
- d) Standing increases carotid sinus stretch and baroreceptor firing, causing reflex bradycardia.

Q73. Which of the following causes the heart to stop in systole?

- a) Hyperkalemia
- b) Hypercalcemia
- c) Hyponatremia
- d) Hypercapnia

Q74. Which of the following is the correct sequence of the spread of depolarization in the ventricles?



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- a) Septum, endocardium, epicardium
- b) Septum, epicardium, endocardium
- c) Endocardium, septum, epicardium
- d) Endocardium, epicardium, septum

Q75. A 31-year-old man is stabbed on the back at the right T10 level. After spinal shock improves, he has right-leg weakness below the lesion. Right-leg vibration and joint-position sense are reduced. Pain and temperature sensation are impaired on the left beginning below the lesion. The examiner asks why different modalities are lost on different sides. Which of the following statements is incorrect?

- a) Right corticospinal tract injury below the pyramidal decussation produces ipsilateral UMN signs below the lesion.
- b) Right dorsal column injury produces ipsilateral loss of vibration and conscious proprioception below the lesion.
- c) Spinothalamic pain and temperature fibers cross near the spinal entry level.
- d) Pain and temperature fibers ascend uncrossed to the medulla before decussating, like dorsal column fibers.

Q76. Depolarization phase of the action potential is due Ca^{2+} influx in which of the following?

- a) Paraspinal muscle
- b) Cardiac muscle
- c) Extraocular muscle
- d) Vascular muscle

Q77. Afferent fibres from baroreceptors go to which of the following?

- a) Nucleus ambiguus
- b) Rostral-ventro lateral medulla
- c) Caudal-ventral lateral medulla
- d) Nucleus of tractus solitarius

Q78. Left shift of the oxygen hemoglobin dissociation curve is seen in which of the following conditions?

- a) Increased PCO_2



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- b) Increased 2,3-BPG
- c) Increased pH
- d) Increased temperature

Q79. A 6-year-old child is brought by his mother with complaint of weakness and difficulty in walking from childhood. She also complains that the child is not able to climb stairs. She also states that he is not able to do his day-to-day activities. Upon inspection, his calves are bilaterally hypertrophied. Absence of which of the following proteins can be the cause of the child's symptoms?

- a) Dystrophin
- b) Myosin
- c) Desmin
- d) Actin

Q80. LH receptors are present on which of the following cells?

- a) Leydig cells
- b) Sertoli cells
- c) Primary spermatocytes
- d) Spermatogonia

Q81. A 12-year-old boy presents with a 4-month history of diminished vision and diplopia. He also experiences tiredness towards the end of the day. On examination, the patient has ptosis of the left eye that improves after a period of sleep. Testing reveals the presence of anti-acetylcholine antibodies in the plasma, a normal thyroid function test and a normal CT scan of the brain and orbit. What is the initial diagnosis?

- a) Grave's disease
- b) Hashimoto's thyroiditis
- c) Myasthenia gravis
- d) Astrocytoma

Q82. Which of the following proteins contributes the maximum to the oncotic pressure?

- a) Albumin
- b) Globulin



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- c) Fibrinogen
- d) Prothrombin

Q83. Which of the following has the maximum clearance?

- a) Inulin
- b) Glucose
- c) Urea
- d) Creatinine

Q84. Positive feedback of which hormone of the following hormones is responsible for the LH surge just before ovulation?

- a) Oestrogen
- b) FSH
- c) ACTH
- d) Progesterone

Q85. Absent 'a' wave of JVP is seen in which of the following conditions?

- a) Decreased right ventricular compliance
- b) Tricuspid stenosis
- c) Junctional rhythm
- d) Atrial fibrillation

Q86. Sudden increase in the dietary intake of potassium causes which of the following?

- a) Increased potassium reabsorption in PCT
- b) Increased potassium secretion in PCT
- c) Increased potassium reabsorption in collecting duct
- d) Increased potassium secretion in collecting duct

Q87. A 69-year-old man is examined after a stroke. Tapping the patellar tendon produces brisk knee extension. The quadriceps contracts while hamstring activity is inhibited. The neurologist explains that tendon tapping stretches the muscle stimulating the muscle spindle. Sensory fibers from the muscle spindle enter the spinal cord and excite the homonymous alpha motor neuron. A



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separate inhibitory interneuron reduces antagonist activity. The student confuses this with inverse myotatic reflex physiology. Which ONE statement is false?

- a) Muscle spindles detect muscle stretch and contribute to the deep tendon reflex.
- b) Ia afferents monosynaptically excite alpha motor neurons of the stretched muscle.
- c) Reciprocal inhibition reduces antagonist muscle contraction during the reflex.
- d) Golgi tendon organ Ib afferents are the primary afferents for the classic patellar stretch reflex.

Q88. Which of the following statements true about GFR?

- a) Afferent arteriolar constriction increases GFR
- b) Afferent arteriolar dilatation decreases GFR
- c) Efferent arteriolar constriction increases GFR
- d) Efferent arteriolar dilatation increases GFR

Q89. Which wave of the JVP corresponds to the third heart sound?

- a) 'a' wave
- b) 'c' wave
- c) 'x' descent
- d) 'y' descent

Q90. Insulin dependent GLUT-4 transport is which of the following methods?

- a) Primary active Transport
- b) Secondary active Transport
- c) Simple diffusion
- d) Facilitated diffusion

Q91. On uterine contraction, there is cervical dilation and delivery. Which of the following shows a similar mechanism?

- a) Milk ejection by Oxytocin
- b) Water retention by ADH
- c) FSH secretion by Anterior Pituitary
- d) T4 secretion by thyroid gland



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- Q92. Lesion of which of the following nuclei of the basal ganglia causes athetosis?
- a) Caudate nucleus
 - b) Globus pallidus
 - c) Subthalamic nucleus of Lewy's
 - d) Substantia nigra pars compacta
- Q93. You are performing a CNS examination on a patient with a stroke. With his eyes closed, you move his big toe up or down while holding his metatarsophalangeal joint. You ask him to state the position of the digit. Which of the following fibers is responsible for the transmission of the sensation tested here?
- a) A alpha
 - b) A gamma
 - c) A beta
 - d) A delta
- Q94. A 60-year-old man is taken to the physician because of a tremor in his hands, trouble sleeping, constipation, and dizziness. Physical examination shows a resting tremor, rigidity and bradykinesia. The man is alert, engaging and optimistic. He speaks in a soft, low voice. Which of the following is likely to be decreased in this patient?
- a) Serotonin in raphe magnus nucleus
 - b) Dopamine in substantia nigra pars compacta
 - c) GABA in substantia nigra pars reticularis
 - d) Acetylcholine in locus ceruleus
- Q95. A right sided subdural hematoma develops in a 30-year-old man after a vehicular accident. Physical examination reveals papilloedema and raised intracranial pressure. Collapse of which of the following structures is the most likely the cause of a decreased oxygenation of the brain in this patient?
- a) Arteries
 - b) Capillaries
 - c) Veins
 - d) Lateral ventricles



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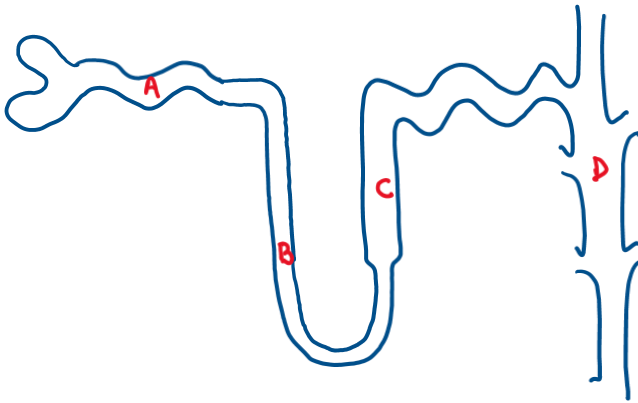
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- Q96. Bilateral ablation of the lateral hypothalamus will lead to which of the following deficits?
- Decreased eating and drinking
 - Loss of sexual drive
 - Excessive eating, rage and aggression, hyperactivity
 - Inability to maintain normal body temperature
- Q97. A 10-year-old boy jumps a flight of stairs and lands on the balls of his feet. The increase in muscle tension causes a sudden, complete relaxation of the affected muscles. Which of the following sensory receptors is most likely to mediate this relaxation of muscles when the tension is increased?
- Muscle spindle
 - Golgi tendon organ
 - Nociceptors
 - Meissner's corpuscle
- Q98. A 53-year-old singer was performing on a hot summer day in a hall with no air conditioning and with the number of attendees being three times the capacity. He was sweating profusely but did not drink fluids. He is brought to the Emergency with a fever of 102 degrees F, heart rate of 140 beats/ min and a blood pressure of 100/55 mm Hg in the supine position. Upon examination he has no perspiration, appears to have dry mucus membranes, and is poorly oriented to person, place and time. Assuming his kidneys were normal, all of the following hormone changes will be expected in this patient except?
- High ADH
 - High Renin
 - High Aldosterone
 - High ANP
- Q99. A 20-year-old male is involved in a vehicular accident. He loses a considerable amount of blood. Which of the following changes are most likely in this patient following the blood loss?
- Increased firing rate of baroreceptors
 - Increased sympathetic discharge
 - Increased Angiotensin II secretion by the JG cells
 - Increased secretion of ANP



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Q100. A marathon runner is running a marathon on a hot, humid day. He has profuse sweating resulting in maximum antidiuresis. Maximum water reabsorption will occur in which of the following segments of the nephron?



- a) A
- b) B
- c) C
- d) D

Q101. Which of the following is an inappropriate hypophyseal hormonal response to the hypothalamic hormone listed?

- a) Somatostatin ↓GH
- b) Dopamine ↑Prolactin
- c) GnRH ↑LH
- d) CRH ↑ACTH

Q102. Which of the following transport processes across the cell membranes is incorrectly described?

- a) Glucose uptake into skeletal muscle is by facilitated diffusion
- b) Transport of glucose from the intestinal lumen into the enterocyte is by secondary active cotransport
- c) Movement of Na^+ ions into a nerve cell during the upstroke of an action potential is by facilitated diffusion
- d) Ouabain-sensitive transport of Na^+ ions from the ICF to ECF is by primary active transport

Q103. All of the following statements are true about amylin except?



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- a) It is secreted by beta cells of pancreas
- b) Ratio of insulin and amylin is 100:1
- c) It enhances gastric emptying
- d) It inhibits glucagon secretion

Q104. Which of the following will cause a decrease in plasma potassium levels by causing a shift in potassium from ECF to ICF?

- a) Strenuous exercise
- b) Aldosterone deficiency
- c) Insulin excess
- d) Beta- adrenergic blockade

Q105. The flexor withdrawal reflex is initiated by stimulation of which of the following receptors?

- a) Muscle spindle
- b) Golgi tendon organ
- c) Joint capsule receptor
- d) Cutaneous free nerve endings

Q106. A high potassium diet is most likely to produce which of the following changes?

- a) Increased secretion of DHEA
- b) Increased secretion of ACTH
- c) Increased secretion of cortisol
- d) Increased secretion of aldosterone

Q107. Migrating motor complex which occur in the inter digestive period is triggered by which of the following?

- a) Motilin
- b) NO
- c) CCK
- d) Somatostatin



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- Q108. A patient is administered sufficient T4 to increase the plasma level of the hormone severalfold. Which of the following changes is the least likely in this patient after several weeks of T4 administration?
- a) Increased heart rate
 - b) Increased cardiac output
 - c) Increased plasma cholesterol
 - d) Increased metabolic rate
- Q109. Acidification of the urine occurs in which of the following segments of the nephron?
- a) PCT
 - b) Loop of henle
 - c) DCT
 - d) CD
- Q110. Which of the following hormones does not need a second messenger?
- a) FSH
 - b) ADH
 - c) Insulin
 - d) Glucagon
- Q111. Which of the following tracts innervates the A gamma motor neurons?
- a) Lateral corticospinal tract
 - b) Vestibulospinal tract
 - c) Reticulospinal tract
 - d) Rubrospinal tract
- Q112. Insulin dependent GLUT 4 transport is by which of the following transport processes?
- a) Primary active Transport
 - b) Secondary active Transport
 - c) Simple diffusion
 - d) Facilitated diffusion



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- Q113. Maximum velocity of blood flow is seen in which of the following blood vessels?
- a) Aorta
 - b) Arteries
 - c) Arterioles
 - d) Capillaries
- Q114. Which of the following describes the phenomenon in RBCs where bicarbonate ions are exchanged for chloride ions to maintain electrical neutrality?
- a) Hamburger effect (Chloride shift)
 - b) Bohr effect
 - c) Haldane effect
 - d) Double Bohr's effect
- Q115. Which of the following is the correct sequence of actin- myosin cross bridge cycling?
- a) Detachment of myosin head, power stroke, hydrolysis of ATP, re-cocking of myosin head
 - b) Hydrolysis of ATP, power stroke, detachment of myosin head, re-cocking of myosin head
 - c) Power stroke, detachment of myosin head, re-cocking of myosin head, hydrolysis of ATP
 - d) Power stroke, detachment of myosin head, hydrolysis of ATP, re-cocking of myosin head
- Q116. Respiratory distress syndrome is due to defect in which of the following?
- a) Type 1 pneumocytes
 - b) Type 2 pneumocytes
 - c) Alveolar capillary endothelial cells
 - d) Bronchial mucosal epithelial cells
- Q117. An individual sleeps with his arm under his head. He wakes up with numbness, tingling and pain in his arm . Which of the following nerve fibres are likely to be affected?
- a) Type A delta
 - b) Type B
 - c) Type C
 - d) Post ganglionic sympathetic nerve fibre



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Q118. Which of the following electrolyte imbalances results in increased nerve excitability?

- a) Hyponatremia
- b) Hypokalemia
- c) Hypocalcemia
- d) Hypochloremia

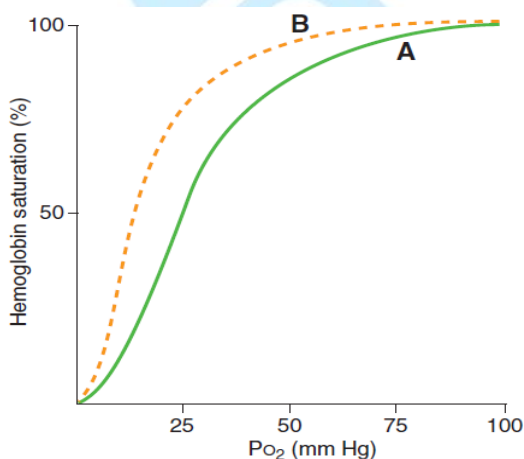
Q119. Increase in which of the following ion in the ECF has the maximum effect on neuronal excitability?

- a) Cl^-
- b) Ca^{2+}
- c) Na^+
- d) K^+

Q120. A patient with HbA_{1c} of 11.4% has come to your clinic with severe pain in his feet at night. Which nerve fibers are involved in this patient?

- a) $\text{A}\alpha$
- b) $\text{A}\beta$
- c) Type B
- d) Type C

Q121. The diagram shown below is the oxygen-haemoglobin dissociation curve. What is the P₅₀ of the oxy-hemoglobin curve labeled 'A' in the given diagram?



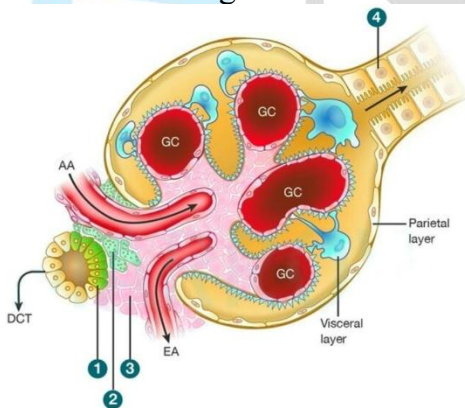
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- a) 5 mm Hg
- b) 25 mm Hg
- c) 50 mm Hg
- d) 75 mm Hg

Q122. V/Q ratio is highest in which part of the lung?

- a) Apex
- b) Base
- c) Middle
- d) None

Q123. A physiologist studies about the sensor which detects elevated sodium and decreases the glomerular filtration rate. In the given image of the juxtaglomerular apparatus, which of the following structures acts as this sensor?



- a) 1
- b) 2
- c) 3
- d) 4

Q124. A patient presents to the emergency department with unconscious state and an arterial blood gas analysis was done. The ABG report is as follows:- Ph = 7.1 Pco₂ = 23 Po₂ = 98 Hco₃⁻ = 16. The most likely condition in the patient is which of the following?

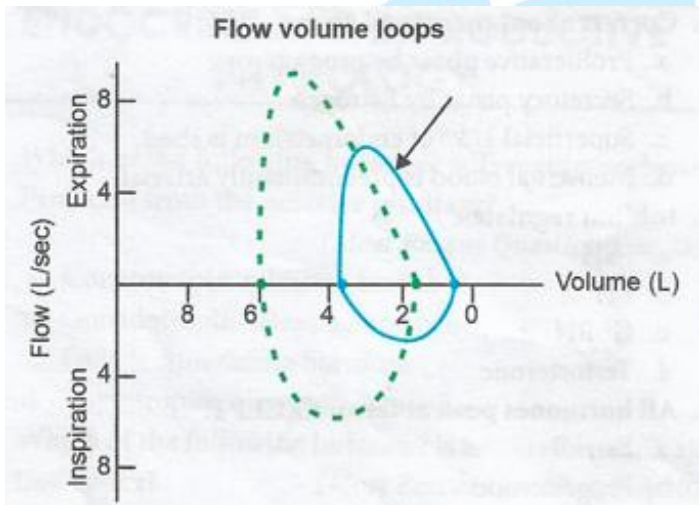
- a) Metabolic acidosis
- b) Metabolic alkalosis



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- c) Respiratory acidosis
- d) Respiratory alkalosis

Q125. A 60-year-old patient complains of breathlessness on climbing stairs. On auscultation heart sounds are normal and normal breath sounds are heard in bilateral lung fields. Flow volume loop is shown/marked in color blue. Comment on diagnosis?



- a) Obstructive airway disease
- b) Parenchymal restrictive disease
- c) Large airway obstruction
- d) Normal report

ANSWERS

1	a
2	b
3	b
4	c
5	b
6	b
7	b
8	d
9	c
10	b
11	c
12	a
13	b
14	c

15	b
16	b
17	c
18	a
19	b
20	c
21	c
22	a
23	a
24	a
25	d
26	d
27	a
28	d

29	d
30	c
31	d
32	a
33	d
34	d
35	b
36	a
37	d
38	d
39	c
40	c
41	a
42	a

43	c
44	a
45	d
46	b
47	d
48	d
49	a
50	b
51	d
52	b
53	a
54	c
55	a
56	b

57	d
58	a
59	d
60	a
61	d
62	d
63	a
64	a
65	b
66	d
67	c
68	d
69	c
70	a



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71	a
72	d
73	b
74	a
75	d
76	d
77	d
78	c
79	a
80	a
81	c

82	a
83	d
84	a
85	d
86	d
87	d
88	c
89	d
90	d
91	a
92	b

93	a
94	b
95	c
96	a
97	b
98	d
99	b
100	a
101	b
102	c
103	c

104	c
105	d
106	d
107	a
108	c
109	d
110	c
111	c
112	d
113	a
114	a

115	d
116	b
117	a
118	c
119	d
120	d
121	b
122	a
123	a
124	a
125	b